Xueji Zou

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Education

University of Kansas Lawrence, KS

Bachelor of Science in Mechanical Engineering; GPA: 4.0/4.0 *Expected May 2029*

Relevant Coursework: Statics, Science of Materials, Differential Equations, MATLAB Programming, Calculus III

Projects

Flying-Wing UAV | *Fusion 360, XFLR5, ANSYS Fluent* Summer 2026

- Reduced drag 38% from the baseline design by iterating a 42 in flying-wing geometry through XFLR5 analysis and ANSYS Fluent CFD.
- Designed a modular 800 g airframe in Fusion 360 with a 3D-printed LW-PLA shell, carbon-fiber spars, and threaded-insert mounts for swappable components.
- Selected and integrated the propulsion system, battery, and GPS-enabled iNav flight controller to target 45+ min endurance and support future autonomous flight.
- Diagnosed CG and launch instabilities from early flight-test failures and built a bungee launcher that produced consistent, low-pitch launches.

Low-Speed Open-Circuit Wind Tunnel | *SolidWorks, Arduino, C++, MATLAB* Spring 2026

- Designed and built a wind tunnel with a $9 \times 9 \times 18$ in test section, using a 3D-printed contraction, honeycomb, and mesh screens to condition the flow.
- Developed an Arduino data acquisition system with a pitot-static tube and differential pressure sensor to measure airspeed up to 32 mph.
- Designed and calibrated a 2-axis load-cell force balance to measure lift and drag across multiple angles of attack.
- Compared measured C_L and C_D to CFD predictions in MATLAB and traced discrepancies to Reynolds-number mismatch and tunnel blockage.

Mousetrap Car | *SolidWorks, Mechanical Design, Prototyping* Fall 2025

- Designed and built a mousetrap-powered vehicle in SolidWorks to travel along a narrow beam and stop near a target location.
- Improved stability and stopping accuracy by testing wheel alignment, axle configurations, and lever-arm lengths.

Experience

Host & Cashier February 2025 – Present

JoyWok *Overland Park, KS*

- Greet and seat guests, manage the waitlist, and pace table assignments across days of up to 450 tickets and 1,300+ guests, including dinner rushes of up to 200 tickets.
- Process cashier transactions and takeout orders accurately, handling cash and card payments at the register.
- Resolve customer concerns and coordinate with servers and kitchen staff to ensure guests are seated, served, and checked out efficiently.

Activities & Honors

Jayhawk Aero Design | University of Kansas Fall 2026 – Present

American Society of Mechanical Engineers (ASME) | University of Kansas 2025 – 2026

Chem-E-Car Competition Team | University of Kansas Fall 2025

Dean's List | University of Kansas 2025 – 2026

Technical Skills

CAD and Analysis: SolidWorks, Fusion 360, Siemens NX, XFLR5, ANSYS Fluent, STAR-CCM+, Revit
Programming and Controls: MATLAB, C++, Arduino IDE, ESP32, iNav